

MASTER OF COMPUTER APPLICATION QUESTION BANK

SUBJECT : 24MCCCST204 – CRYPTOGRAPHY AND CYBER SECURITY
SEM/YEAR: II/I

Module I

Introduction to Cryptography, OSI security architecture: Security Services, Mechanisms and Attacks- Phishing, Ransomware, DoS attack. Network security model. Classical Encryption Techniques - Symmetric cipher model, substitution techniques, and transposition techniques. Steganography.

Q No	Question	BTL	Competence
1.	Define Cryptography.	1	Remember
2.	Outline security services and mechanisms in cryptography	1	Remember
3.	Describe different various security services in detail	1	Remember
4.	List the four categories of security threats.	1	Remember
5.	Explain the difference between Phishing & Ransomware Attacks	1	Remember
6.	What do you understand by network security attacks? Describe active and passive security attacks.	1	Remember
7.	Explain which encryption is suitable for data transmission	1	Remember
8.	Describe symmetric cipher model in cryptography	1	Remember
9.	Differentiate symmetric and asymmetric encryption?	1	Remember
10.	Outline any two poly-alphabetic substitution techniques with suitable examples	1	Remember
11.	Give any four types of substitution techniques.	1	Remember
12.	Identify and define with example two classical encryption Techniques?	1	Remember
13.	Given the key “MONARCHY”, apply the Playfair cipher to the Plaintext “FACTIONAISM”. Decrypt the ciphertext also.	1	Remember
14.	Define play fair cipher. Discuss its encryption, decryption using the following data Plain Text: A lifetime of training for just ten seconds. KEY : Gates	1	Remember
15.	Outline Substitution and Transposition techniques.	1	Remember
16.	Define Diffusion & Confusion.	1	Remember
17.	Convert “MEET ME” using Hill cipher with the key matrix. Convert the cipher text back to plaintext.	1	Remember
18.	Encrypt the message “PAY” using Hill cipher with the following key matrix and show the Encryption to get the Cipher text $K = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$	3	Apply
19.	Describe steganography method with text covering process	1	Remember

Module II

Conventional Symmetric Key Encryption: Block ciphers and Stream Ciphers, Block Cipher Design Principles, Modes of operation, Data Encryption Standard, Advanced Encryption Standard (AES), Multiple Encryption and Triple DES. Public key cryptography: Principles of public key cryptosystems-The RSA algorithm-Key management – Diffie Hellman Key exchange - Elliptic curve arithmetic - Elliptic curve cryptography

Q No	Question	BTL	Competence
1.	Discuss difference between a block cipher and a stream cipher?	2	Understand
2.	Classify advantages and disadvantages of Cipher Feed Back mode & Output Feed Back mode.	2	Understand
3.	Briefly explain the design principles of block cipher	2	Understand
4.	With a suitable diagram explain Cipher Block Chaining (CBC) mode.	2	Understand
5.	Write short notes on Feistel cipher structure	2	Understand
6.	Discuss the important to study Feistel cipher.	2	Understand
7.	Explain Data Encryption Standard with a neat diagram.	2	Understand
8.	Explain triple DES encryption and Decryption? Explain the term meet-in-the middle attack	2	Understand
9.	Discuss the strength and weaknesses of DES.	2	Understand
10.	Explain Avalanche effect.	2	Understand
11.	Explain the Key generation process in Data Encryption Standard (DES) algorithm.	2	Understand
12.	Draw the general structure of DES and explain the encryption decryption process.	2	Understand
13.	Explain AES algorithm in with a neat diagram.	2	Understand
14.	Using AES, perform the first round transformation for the following: Plain text(128 bit):00112233445566778899aabbccddeeff, key(128-bit): 000102030405060708090a0b0c0d0e0f, Represent them as 4x4 matrix and explain the first round	3	Apply
15.	With neat diagram explain the 4 main transformations in AES algorithm.	2	Understand
16.	With a neat diagram explain encryption and authentication using public key cryptography	2	Understand
17.	Define Asymmetric key Cryptography and explain about the RSA cryptosystem with suitable example.	1	Remember
18.	Explain RSA algorithm in detail with an example?	2	Understand
19.	Explain Diffie-Hellman key exchange algorithm with suitable diagrams	2	Understand
20.	For a Deffie-Hellman scheme with common prime $q = 11$, and primitive root $= 2$, generate a set of public key and private key pairs between two users A and B. Make your own assumptions.	2	Understand
21.	Consider a Diffie-Hellman scheme with a common prime $q = 11$ and a primitive root $a = 2$. Show that 2 is a primitive root of 11 If user A has public key $Y_A = 9$, what is A's private key X_A ? If user B has public key $Y_B = 3$, what is the shared secret key K ?	2	Understand
22.	Explain the RSA algorithm and discuss the timing attacks on RSA algorithm. Let $p=7$, $q=11$, $e=17$, $M=8$ perform encryption and decryption using the RSA algorithm	2	Understand
23.	For a Diffie-Hellman key exchange, given the public values $p=23, g=5$ and primitive keys $a=6, b=15$, Compute the shared key	3	Apply

Module III

Hash Functions and MAC: Properties of hash functions, birthday attack, hashcash, Message Authentication Code Algorithms, MAC protocols: HMAC, CMAC. Digital Signatures: Classification of signature schemes: RSA signature, Digital Signature Standard, Overview of ElGamal and Schnorr schemes, One time signature schemes, Attacks on Digital Signatures, Blind Signatures.

Q No	Question	BTL	Competence
1.	Discuss the requirements for message authentication?	2	Understand
2.	Define hash functions? What is its significance?	1	Remember
3.	Discuss about what do you understand by message authentication code? How it differs from one way hash function?	2	Understand
4.	Write a note on Birthday Attack	1	Remember
5.	Write a note on Hash Cash	1	Remember
6.	Discuss about the objectives of HMAC and its security features	2	Understand
7.	Define the concept of Message Authentication and Explain Cipher-Based Message Authentication code (CMAC) in detail	1	Remember
8.	Describe various attacks on digital signature	1	Remember
9.	List out the classification of digital signature schemes.	1	Remember
10.	Explain RSA Digital Signature scheme and attacks on RSA Digital Signature schemes	2	Understand
11.	Discuss clearly Secure Hash Algorithm(SHA)	2	Understand
12.	Explain the concept of Blind Signatures. Discuss its working, applications, and security considerations with suitable examples	2	Understand
13.	Illustrate Blind Signature scheme in detail	3	Apply
14.	Discuss One-Time signature schemes	2	Understand
15.	Explain the ElGamal Digital Signature Scheme	2	Understand
16.	Explain the Schnorr Digital Signature Scheme	2	Understand
17.	Illustrate how MAC is generated and verified using hmac with the given example: Message = "Hello", secret key=K	3	Apply
18.	Illustrate the process of generating and verifying a Message Authentication Code (MAC) using HMAC for the following: • Message: "SecureData" • Secret Key: "Key123" Explain clearly: 1. Key preparation 2. Inner hash computation 3. Outer hash computation 4. MAC verification at the receiver side	3	Apply

Module IV

Introduction to Cyber Security: Email Security: Security Services for email, Attacks possible through email, Establishing keys privacy, authentication of the source, Message Integrity, Non- repudiation, Pretty Good Privacy, S/MIME. IP Security: Overview of IPSec, IPv4 and Ipv6, Authentication Header, Encapsulation Security Payload (ESP), Internet Key Exchange. Transport Level Security: SSL/TLS Basic Protocol, computing the keys, client authentication, PKI as deployed by SSL, Attacks fixed in v3, Exportability, Encoding, Secure Electronic Transaction (SET).

Q No	Question	BTL	Competence
1.	Define Non-Repudiation? Explain Non-Repudiation based on Public Key Technology.	2	Understand
2.	List out major Email security challenges	1	Remember
3.	Explain security services for Email.	2	Understand
4.	Explain the architecture of S/MIME	2	Understand
5.	Explain Object formats and Primitive Object formats in PGP	2	Understand
6.	Explain the operational description of PGP.	2	Understand
7.	Define Pretty Good privacy (PGP). Give the structure of Pretty Good privacy? Also explain the concept of key rings with their formats	2	Understand
8.	Illustrate PGP message format in Email security.	3	Apply
9.	Explain Exportability in SSLv2 and Exportability in SSLv3	2	Understand
10.	Distinguish between IPV4 & IPV6		
11.	Explain two modes of IPSEC	2	Understand
12.	Outline the IPSec Encapsulating Security Payload in detail.	1	Remember
13.	Describe Encapsulating Security Payload (ESP).	1	Remember
14.	Write short notes on authentication header and ESP.	1	Remember
15.	Differentiate transport and tunnel mode in IPSec	2	Understand
16.	Explain about PKI in detail.	2	Understand
17.	Discuss the protocols of SSL.	2	Understand
18.	Describe the working of secure electronic transaction with suitable diagrams	1	Remember
19.	List out the features of SET.	1	Remember
20.	Describe the major security aspects in the security of electronic mail system.	2	Understand
21.	Discuss about What services does IPSec provide? What is the difference between transport mode and tunnel mode?	2	Understand
22.	Summarize the SSL/TLS Handshake Protocol	2	Understand

Module V

Cyber Crimes and Cyber Security: Cyber Crime and Information Security – classifications of Cyber Crimes – Tools and Methods – Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control–Cloud Security – Web Security – Wireless Security

Q No	Question	BTL	Competence
1.	Write short note on a) web security b) wireless security	1	Remember
2.	Discuss the tools and methods used in cybercrimes.	2	Understand
3.	Classify and analyse the various types of cybercrimes	2	Understand
4.	Discuss SQL injection	2	Understand
5.	Discuss the fundamental principles of cyber security	2	Understand
6.	Discuss the core principles of information security	2	Understand
7.	Discuss Network Access control based on cyber security	2	Understand
8.	Discuss Cloud Security	2	Understand